New Attempt

- Due Nov 6, 2024 by 2:59am
- Points 10
- Submitting a text entry box, a website url, or a file upload

Robot Project

Robot Project Challenge 1

Overview:

This is your first challenge as a team. In this challenge you will meet and discuss goals, responsibilities, communication and expectations for your team. Then AS A TEAM you will build and program the robot to drive in a straight line, staying within a path, from end to end on the table and stop. You should record all the steps and documents in your digital design notebook.

Reminder: It's crucial to dedicate time to your individual lasercutting project. Afterward, incorporate the best design of your team into your robot, ideally by **Challenge 2.**

Project Teams and Indicating your free times

1. Please assign yourself to one of the predefined teams by navigating to the 'People' menu and selecting the 'Robot Project' tab.

2. Please complete this **When2Meet** **(https://www.when2meet.com/?26974475-FSBWk)** scheduling calendar to indicate your free times during the weekdays and on Saturdays, in general, for in-person work.Prof. Keyvani will use this information to determine the optimal times for her in-person and virtual office hours, as well as the maker classroom operating hours, aiming to accommodate the schedules of as many students as possible.

Please note that, you'll need to dedicate time to this project, so if your schedule is already full, it's important to make adjustments and prioritize accordingly.

- ## Tasks:
  - **Task 1: Create a team Charter and add this to your digital design notebook. As a TEAM, create a single document (~2 pages max) signed by all of you that answers the**

following questions. Your team charter will serve as a reference for conflict resolution within the team. Use the following template:

- **All about your team:**
  - List the followings for each team member:
    - Name
    - Email
    - Phone Number
  - Identify the method by which you will primarily communicate with each other
- **What are your team's norms? norms are the specific, often unwritten, rules or expectations that guide the behavior of team members within the team.** Provide specific examples or scenarios that highlight how these norms guide your team's interactions and decision-making processes. Example: Punctuality, active participation in meetings, expectations of the work quality and how conflicts are resolved, add your own.
- **What are your team's core values? values are the shared beliefs, principles, or standards that the team collectively upholds and considers important.** Don't just make a list of your team values; explain how you would follow and support them in your team. Offer examples or scenarios for clarification. Examples of core values: Respect, Fairness, Commitment, Transparency, Inclusion, add your own.
- **Team Charter violation :** In the event of a team member violating established norms and values, your team should adhere to a fair and transparent process. **Please outline your team's process and consequences here.**
  - In the event of team issues, this section of the team charter will serve as a reference for guidance.
  - In extreme situations where a team member consistently fails to meet established norms and values, and all efforts to address and rectify the situation, including notification and discussion, correction Period and involvement of the Professor have been exhausted, the team may consider terminating the team member's involvement. In such a scenario, the affected team member has the option to join another team of their choice or continue the project independently.

- **Task 2**: **Use this instruction to assemble your robot (https://northeastern.instructure.com/courses/189179/files/30516330?wrap=1) (https://northeastern.instructure.com/courses/189179/files/30516330/download?download_frd=1) . You can find all the necessary material in the maker classroom.**
  - Remember that you'd need your Redboard and Breadboard to complete your SF homework assignments as well.
  - Give yourself double the amount of time you think would take to assemble.
  - Start early to allow yourself time to fail and fix.
  - Take lots of pictures from your assembly process and include in your digital design notebook.

        - You should update the digital design notebook every time you go the maker classroom to assemble your robot.
    - **Task 3: Program the robot to drive in a straight line, staying within a path, from end to end and stop.**
        - Your first robot task is to have your robot move in a straight line across the table (4 feet) from one point to another and stop.
        - The path is 4 feet long and 2 feet wide.
        - Take what you learned in Project 5A and look at the code in Projects 5B and 5C and think about how you will program your robot to do this.
        - Flowchart your algorithm for programming the robot using draw.io and submit
        - Record a video of your best attempt (can be from during class or out of class)
            - You can include a link to the video on YouTube, Google drive, or other cloud service if you can't/don't want to upload the actual video
        - Submit your Arduino Code.
        - Be sure to update your digital design notebook every time you work on your robot project individually or as a team.

    Task 4: Record everything in your digital design notebook and share the link with Professor. Make sure the link is accessible and does not require permission. You can test this by copy and pasting the link to an incognito window. Use this template (https://docs.google.com/document/d/1bPI3x9p760cnOWCTysXDKUcULuhJ_Y1UThksSpdYNIY/co and address the followings:
        - Your team charter
        - Problem definition
        - Flowcharts
        - Plenty of pictures
        - Meeting minutes
        - Any success or failure
        - A link to a video of your best attempt

File Submission

As a group, submit the followings:

- Provide an accessible link to your digital design notebook. Your notebook should have the following the least:
    - Your team charter
    - Problem definition
    - Flowcharts
    - Plenty of pictures
    - Meeting minutes

- Any success or failure
- A link to a video of your best attempt
- Arduino code

<table>
<tr><th colspan="7">Robot Challenge 1</th></tr>
<tr><th>Criteria</th><th colspan="5">Ratings</th><th>Pts</th></tr>
<tr><td>Robot Success (in-person or video)
Requirements: travel 4ft and stop, stay within the line</td><td>3 pts
Full Marks</td><td>2 pts
Reasonable</td><td colspan="2">1 pts
Needs some work</td><td>0 pts
No Marks</td><td>3 pts</td></tr>
<tr><td>Code uploaded and compiles</td><td colspan="2">3 pts
Full Marks</td><td colspan="3">0 pts
Does not compile or has errors</td><td>3 pts</td></tr>
<tr><td>Digital Design Notebook Details
Has all of the details requested</td><td>4 pts
Full Marks</td><td colspan="2">2 pts
Needs some work</td><td colspan="2">0 pts
No Marks</td><td>4 pts</td></tr>
<tr><td colspan="7" align="right">Total Points: 10</td></tr>
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